

Heat Equation

Let u is a dependent variable and x, y, z, t are independent variables then

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{c^2} \frac{\partial u}{\partial t}$$

One dimensional heat equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{1}{c^2} \frac{\partial u}{\partial t}$$

Two dimensional heat equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{1}{c^2} \frac{\partial u}{\partial t}$$

Three dimensional heat equation

Here u is a solution of heat equation.

Let solve One dimensional heat equation by method of separation

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Three dimensional heat Equation

Here u is a solution of heat equation.

We solve One dimensional heat equation by method of separation

of variables.

learn

$$\begin{array}{|c} x=0 \\ u=0 \end{array}$$

$$\begin{array}{|c} x=l \\ u=0 \end{array}$$

We use Boundary conditions to find constant

- (i) $u=0$ when $x=0$ $u(0,t)$
(ii) $u=0$ when $x=l$ $u(l,t)$

Here l is length of rod

Imp find the solution of
One dimension heat equation

By the method of separation of variables solve the PDE

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$$

of variables.

learn

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Ans Solution of one dimensional heat equations

By formula of one dimension heat equation

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{c^2} \frac{\partial u}{\partial t} \quad \text{--- (1)}$$

By method of separation of variables

Let our solution is

$$u = X \cdot T \quad \text{--- (2)}$$

here 'X' is the function of 'x' only and 'T' is the function of 't' only

Since u is a solution so satisfy equation (1) put the value

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$$\frac{\partial^2 (XT)}{\partial x^2} = \frac{1}{c^2} \frac{\partial (XT)}{\partial t}$$

$$T \frac{\partial^2 X}{\partial x^2} = \frac{1}{c^2} \left(X \frac{\partial T}{\partial t} \right)$$

$$T X'' = \frac{1}{c^2} X T'$$

separate variables one side X and
other side T

$$\frac{X''}{X} = \frac{1}{c^2} \frac{T'}{T}$$

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Imp learn

Value may be
+ve or -ve or zero

$+p^2$ or $-p^2$ or 0

We solve all three cases

Remember

$$\frac{\partial^2 (XT)}{\partial x^2} = \frac{1}{c^2} \frac{\partial (XT)}{\partial t}$$

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Remember

Case I

$$\frac{x''}{x} = \frac{1}{c^2} \frac{T'}{T} = +p^2 \text{ (let)}$$

$$\frac{x''}{x} = p^2$$

$$x'' = p^2 x$$

$$x'' - p^2 x = 0$$

by C.F & P.I. method

$$\text{A.E. } m^2 - p^2 = 0$$

$$m^2 = p^2$$

$$m = +p, -p$$

$$\text{C.F.} = \zeta_1 e^{px} + \zeta_2 e^{-px}$$

$$\text{P.I.} = 0$$

$$x = \zeta_1 e^{px} + \zeta_2 e^{-px}$$

$$\frac{1}{c^2} \frac{T'}{T} = p^2$$

$$T' = c^2 p^2 T$$

$$T' - c^2 p^2 T = 0$$

by Cof & P.O.I. method

$$\text{A.E. } m^2 - c^2 p^2 = 0$$

$$m = \pm c p$$

$$\text{C.F.} = \zeta_3 e^{c p^2 t}$$

$$\text{P.I.} = 0$$

$$c^2 p^2 t$$

$$T = \zeta_3 e^{c^2 p^2 t}$$

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$$c^2 p^2 t$$

$$T = \zeta e^{c^2 p^2 t}$$

put the value in (2)

In this case our heat equation
Solution is

$$u = X T$$

$$u = \left(C_1 e^{px} + C_2 e^{-px} \right) \int_0^x e^{c^2 p^2 t} dt \quad \text{--- (3)}$$

Very important case
Case II : Suitable solution
आइए Heat Wave Laplace नीचे
लिए

$$\frac{x''}{x} = \frac{1}{c^2} \frac{T'}{T} = -p^2 \text{ (let)}$$

$$\frac{x''}{x} = -p^2$$

$$x'' = -p^2 x$$

$$x'' + p^2 x = 0$$

by C.F & P.I. method

$$\text{A.E. } m^2 + p^2 = 0$$
$$m^2 = -p^2$$

$$\frac{1}{c^2} \frac{T'}{T} = -p^2$$

$$T' = -c^2 p^2 T$$

$$T' + c^2 p^2 T = 0$$

by C.F and P.I. method

put the value in (2)

In this case our heat equation
solution is

$$u = XT$$

$$u = \left(C_1 e^{px} + C_2 e^{-px} \right) \int_0^t e^{-c^2 p^2 t} dt \quad \text{--- (3)}$$

Very important case
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A. E. & Heat Wave Laplace method
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$$T' + c^2 p^2 T = 0$$

by C.F. and P.I. method

$$X'' + p^2 X = 0$$

$$\left(\frac{d^2}{dx^2} + p^2 \right) X = 0$$

A.E. $m^2 + p^2 = 0$

$$m^2 = -p^2$$

$$m^2 = i^2 p^2$$

$$m = \pm ip$$

$$Cof_0 = C_1 \cos px +$$

$$C_2 \sin px$$

$$P.I = 0$$

$$X = Cof + P.I.$$

$$X = C_1 \cos px +$$

$$C_2 \sin px$$

$$T' + c^2 p^2 T = 0$$

$$\left(\frac{d}{dt} + c^2 p^2 \right) T = 0$$

A.E. $m + c^2 p^2 = 0$

$$m = -c^2 p^2$$

$$Cof = C_3 e^{-c^2 p^2 t}$$

$$P.I = 0$$

$$T = Cof + P.I$$

$$T = C_3 e^{-c^2 p^2 t}$$

put the value in equation (2)

In this case our heat equation solution is

$$u = X \cdot T$$

Learn

$$u = (C_1 \cos px + C_2 \sin px) \cdot C_3 e^{-c^2 p^2 t}$$

$$X'' + p^2 X = 0$$

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$$X = C_1 \cos px +$$

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$$T' + 2p^2 T = 0$$

$$\left(\frac{d}{dt} + 2p^2 \right) T = 0$$

A.E. $m + 2p^2 = 0$

$$m = -2p^2$$

$$-c^2 p^2 t$$

$$Cof = C_3 e$$

$$P.I = 0$$

$$T = Cof + P.I$$

$$T = C_3 e^{-c^2 p^2 t}$$

put the value in equation (2)

In this case our heat equation solution is

$$u = X \cdot T$$

$$u = (C_1 \cos px + C_2 \sin px) \cdot C_3 e^{-c^2 p^2 t}$$

Learn

Case III

$$\frac{X''}{X} = \frac{1}{c^2} \frac{T'}{T} = 0 \text{ (let)}$$

$$\frac{X''}{X} = 0$$

$$X'' = 0$$

by C.F & P.I method

$$A.E. \quad m^2 = 0$$

$$m = 0, 0$$

repeated roots

$$C.F = C_1 + x C_2$$

$$P.I = 0$$

$$X = C_1 + x C_2$$

$$\frac{1}{c^2} \frac{T'}{T} = 0$$

$$T' = 0$$

$$\frac{dT}{dt} = 0$$

by C.F & P.I method

$$A.E. \quad m = 0$$

or

$$C.F = C_3 e$$

$$= C_3$$

$$P.I = 0$$

$$T = C_3$$

put the value in (2)

In this case our heat equation

solution is

$$u = XT$$

Case III

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$$\frac{X''}{X} = 0$$

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by C.F & P.I. method

$$A.E. \dots m^2 = 0$$

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repeated roots

$$C.F = C_1 + x C_2$$

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$$X = C_1 + x C_2$$

$$\frac{1}{c^2} \frac{T'}{T} = 0$$

$$T' = 0$$

$$\frac{dT}{dt} = 0$$

by C.F & P.I. method

$$A.E. \dots m = 0$$

or

$$C.F = \frac{C}{3} e$$

$$= \frac{C}{3}$$

$$P.I = 0$$

$$T = \frac{C}{3}$$

put the value in (2)

In this case our heat equation

solution is

$$u = XT$$

$$u = XT$$

$$u = (C_1 + x C_2) \cdot C_3 \quad \text{--- (5)}$$

Of these three solutions, we have to choose that solution which is consistent with the physical nature of the problem.

Hence Case II for $-p^2$ is only suitable solution of the heat equation.

$$u = (C_1 \cos px + C_2 \sin px) \cdot C_3 e^{-C_3 p^2 t}$$

Learn

This is the solution of heat equation.

$$u = XT$$

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$$u = (C_1 \cos px + C_2 \sin px) \cdot C_3 e^{-c p^2 t} \quad \text{--- (6)}$$

Learn

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Of these three solutions, we have to choose that solution which is consistent with the physical nature of the problem.

Hence Case II for $-p^2$ is only suitable solution of the heat equation.

$$u = (C_1 \cos px + C_2 \sin px) \cdot C_3 e^{-C_3^2 p^2 t} \quad \text{--- (6)}$$

Lesson

This is the solution of heat equation